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GreenArrays Technology Brief

1 Technology and Engagement

Editorial note. Cartheur Research acquired the full GreenArrays IP stack in 2025.

Cartheur Research combines GreenArrays multicomputer technology with GLOW, its in-house design environment, to develop fine-grained, energy-conscious silicon for problems that benefit from a direct route between system idea and implementation.

1.1 Products and Services

- Chip production and sales.
- Custom design services.
- Collaboration in product development.
- Support and education.

1.2 Why GreenArrays Technology

- Fine-grained parallel computing in compact, low-energy systems.
- A direct connection between system architecture, custom silicon, and application software.
- Custom design assistance for distinctive product constraints.
- Reusable methods, tools, and engineering knowledge across programs.

1.3 Working Principles

- Use resources deliberately.
- Keep design decisions close to the engineers responsible for the outcome.
- Build practical, supportable products.
- Match customer commitment with sustained technical collaboration.

2 GreenArrays Heritage and Current Direction

Historical context. GreenArrays was founded in 2009 to develop multi-computer chips and apply them to demanding engineering problems. Cartheur Research acquired the full GreenArrays IP stack in 2025 and is carrying that technology forward through custom design, product-development collaboration, chip supply, support, and education.

GreenArrays technology was created for engineering teams that need more than an off-the-shelf computing component. Its heritage includes multi-computer chips, custom design work, and practical support for teams turning a distinctive system idea into a product.

People. Charles H. (Chuck) Moore and his long-standing collaborators established the technical foundation for GreenArrays. That foundation joins silicon design, circuits, application software, and design tools in one engineering discipline—a model Cartheur continues through its own team and specialist partners.

Efficiency. The GreenArrays approach begins with Chuck Moore’s creation of Forth in 1968: give practitioners a direct, compact way to express, test, and refine an idea. Decades of work in software and silicon extended that principle from code into computer architecture, favoring the complexity that serves the product and removing the rest.

In the 1980s, Chuck expanded this paradigm by applying first principles to computer design, accepting only the complexity necessary for practical usefulness. At Novix, he developed the NC4000 processor; Harris Semiconductor licensed the design and marketed an enhanced version as the RTX2000. This work led through evolutionary steps to the full-custom silicon designs offered today.

Capital efficiency. GreenArrays developed its own CAD software, GLOW, to keep design control and iteration authority with the people making architectural and silicon decisions. The aim is straightforward: direct engineering effort toward the silicon, validation, and customer problem rather than avoidable tool friction and disconnected hand-offs.

GLOW puts the design environment directly in the hands of the people making architectural and silicon decisions. Rather than accepting a fixed tool chain and its assumptions, the team adapts the environment to the problem at hand. It tests alternatives early, carries intent from architecture into implementation, and revises a design without waiting on external tool vendors or reorganizing a fragmented flow.

The capital benefit is cumulative. An in-house environment reduces the recurring cost and friction of per-designer tool licenses, disconnected hand-offs, format conversion, and retraining around a changing vendor stack. It also keeps design knowledge in the organization: methods, checks, and improvements become reusable assets rather than one-off work. Those shorter feedback loops help a small, cross-trained team concentrate investment on silicon, validation, and the customer problem being solved.

Design-tool lineage. GLOW grew from OKAD (“Oh CAD”), Charles H. Moore’s earlier in-house VLSI layout and simulation environment. A 1995 *Forth Dimensions* article described OKAD as a compact full-custom design system with transistor-level layout, layered geometric views, and electrical simulation of an entire chip. The article reported that its functionality had been verified in the production of high-performance microprocessors and signal-processing devices.

This is a characteristic evidence-based lineage of grassroots innovation. Forth gave a practitioner a direct, compact way to express and test an idea; OKAD carried that same principle into silicon, making layout and simulation part of the designer’s own working environment. GLOW continues that tradition by keeping the tool, the architecture, and the engineering judgment close together.

Long-term view. Enduring products require a long-term commitment to deliver and support them. Cartheur Research is designing AI chips around structural algebra developed in-house. Volatco puts this hardware/software direction to work with two GA144 chips and 288 autonomous F18A nodes, executing concurrent workloads without a central-clock bottleneck and driving ultra-low-energy operation.

In conclusion. GreenArrays multi-computer chips combine fine-grained parallel computing, compact design, and low-energy operation. Cartheur helps customers apply that capability to practical products where architectural control and efficient use of resources matter.

3 How We Work With Partners

Cartheur Research brings research and custom design, production and systems delivery, technical support, and application and education partnerships into one customer-facing capability.

3.1 Research and Custom Design

The research and custom-design team develops silicon and uses it to address application problems through hardware, software, and external-circuit design. Engagements can include product definition, architecture, implementation, and validation for systems built with GreenArrays technology.

3.2 Production and Systems Delivery

Chip production and delivery are being extended through the systems-manufacturing experience at Volatco. This creates a practical path from silicon capability to systems in use.

Cartheur begins these engagements with the system problem rather than a tool choice: the workload, energy envelope, interfaces, physical constraints, and manufacturing assumptions. GLOW keeps the resulting exploration close to the engineers responsible for the design. Customers receive explicit trade-offs and reviewable artifacts, with a partner able to revisit an architectural decision while its application context is still clear. This approach is especially relevant to energy-constrained edge systems, tightly coupled hardware/software products, and practical “AI in your pocket” applications.

3.3 Technical Support

Technical support includes software libraries, documentation, application notes, and training materials that help customers bring a design into use and carry it forward.

3.4 Application and Education Partners

Application partners contribute market expertise while Cartheur contributes technology, prototype development, product planning, design, testing, and chip or technology supply. Educational partnerships extend practical learning in parallel computing and digital silicon design.

The Polysance Initiative is part of this educational direction.

4 Technology Milestones

1985: Novix NC4000P One 8 MIPS CPU, external memory and I/O, and an external clock.

2009: GreenArrays GA144 144 complete 666 MIPS computers, with integrated memory and I/O and no clock.

2022: GreenArrays GA144 version 2.0 Technical and speed improvements, including fast SERDES.

Current development: GA144-v2.5 Design kick-off for a structured-algebra silicon AI chip.

5 Selected Applications



**High-reliability
large-scale networks**



Industrial control



Radiotelescope control

Figure 1: Examples of the application spaces addressed by GreenArrays technology.

6 Contact and Notices

For current information and support, visit <https://cartheur.com>.

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